

# Sanket Kumar

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## SUMMARY

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Computer Science undergraduate with hands-on experience in Backend Development, DevOps, and Artificial Intelligence through academic projects. Strong foundation in Java, Python, REST APIs, and automation, with an interest in building scalable applications.

## EDUCATION

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| <b>Vellore Institute of Technology</b>                       | 2023 – Present    |
| <i>Bachelor of Technology in Computer Science CGPA: 8.69</i> | <i>Vellore</i>    |
| <b>Baldwin Farm Area High School</b>                         | 2023              |
| <i>Class XII 80%</i>                                         | <i>Jamshedpur</i> |
| <b>Baldwin Farm Area High School</b>                         | 2021              |
| <i>Class X 93%</i>                                           | <i>Jamshedpur</i> |

## PROJECTS

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### **Fruit Quality Detection System** | *Python, TensorFlow Lite, OpenCV, Raspberry Pi*

- Designed and deployed a real-time fruit classification system on Raspberry Pi using TensorFlow Lite and OpenCV.
- Integrated an MQ-135 ethylene gas sensor with ADS1115 ADC to classify fruit ripeness as *Fresh*, *Ripening*, or *Overripe*.
- Enabled early detection of spoiled fruits to improve food quality monitoring and reduce supply chain waste.

### **DevOps-Enabled To-Do REST API** | *Java, Jenkins, Docker, Terraform, Ansible*

- Built and tested a RESTful Java API using JUnit with an automated CI/CD pipeline powered by Jenkins.
- Containerized the application with Docker and provisioned infrastructure using Terraform and Ansible.
- Monitored application performance and system health using Graphite and Grafana dashboards.

### **Adaptive Cyber Defense System (Reinforcement Learning)** | *Python, PyTorch, NumPy*

- Developed a multi-adversary cyber defense system using Deep Q-Networks (DQN).
- Simulated evolving attacker behavior and adaptive defense actions (Allow, Block, Throttle, Observe, Deceive) using reinforcement learning.
- Trained the model over 30,000 simulation steps to achieve stable policy convergence.

## TECHNICAL SKILLS

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**Languages:** Python, Java, C, C++, JavaScript, HTML/CSS

**Technologies:** Docker, Jenkins, Terraform, Ansible, Git, GitHub, React.js, Node.js, Django, Angular, TensorFlow, PyTorch, OpenCV, Raspberry Pi, JUnit

**Databases:** Oracle SQL, MySQL, Firebase

**Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Compiler Design, Operating Systems, REST API Development, Artificial Intelligence, DevOps, Cloud Computing

## CERTIFICATIONS

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- **DevOps, Agile & Design Thinking** – IBM Career Education Program *Jul 2026*  
Credential ID: CEDEA1IN — Certificate  
Hands-on exposure to DevOps practices, Agile workflows, Design Thinking, and CI/CD pipelines.
- **DevOps Fundamentals** – IBM Career Education Program *Jun 2025*  
Credential ID: CEDEV2IN — Certificate  
Covered Docker, automation, cloud deployment, infrastructure provisioning, and CI/CD fundamentals.

## LEADERSHIP & EXTRACURRICULARS

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- **Senior Core Member, Geospatial Club – VIT** *Technical Domain*  
Developed and maintained the club website while organizing technical workshops and events.